THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS TWO-CHAPTER TEST SET C

Chapters 9 & 10: **Symmetry, The Other Side of Zero** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. How many lines of symmetry does a rectangle (not a square) have? 1

(a) 0 **(b)** 4 **(c)** 1 **(d)** 2

2. An isosceles triangle has: 1

(a) 0 lines **(b)** 1 line of symmetry

(c) 2 lines **(d)** 3 lines

3. Which letter has both horizontal and vertical lines of symmetry? 1

(a) E **(b)** A **(c)** F **(d)** H

4. The point about which a figure is rotated is the: 1

(a) radius **(b)** vertex of reflection

(c) arm **(d)** centre of rotation

5. The angle of rotational symmetry of an equilateral triangle is: 1

(a) 180° **(b)** 90° **(c)** 120° **(d)** 60°

6. The opposite of +5 is: 1

(a) 0 **(b)** −5 **(c)** 5 **(d)** +10

7. In the token model, a red token represents: 1

(a) +1 **(b)** 0 **(c)** 2 **(d)** −1

8. The smallest integer among −8, −2, 0, 3 is: 1

(a) −8 **(b)** −2 **(c)** 3 **(d)** 0

9. $(-5) + (-3)$ equals: 1

(a) −2 **(b)** 2 **(c)** −8 **(d)** 8

10. $(-3) - (-5)$ equals: 1

(a) −2 **(b)** 2 **(c)** 8 **(d)** −8

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. What is the order of rotational symmetry of the letter H? 2

12. Write the angle for which a pentagon coincides itself on rotation. 2

13. Is zero a positive integer? Give a reason. 2

14. The temperature at night is −3°C and rises by 8°C by noon. What is it at noon? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. What is meant by ‘order of rotational symmetry’? Give one example. 3

16. Find the integer which when added to −9 gives 4. 3

17. State which has greater order: a regular pentagon or a regular hexagon. Why? 3

OR

Explain the difference between line symmetry and rotational symmetry. 3

18. State the signs of the sum of (−12) and 7. Explain. 3

OR

Brahmagupta gave rules about debts and fortunes. Express: a debt of 5 added to a debt of 3. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. **Case study:** A logo is a regular pentagon with a star inside. 5

(i) Lines of symmetry?
(ii) Angle of rotation?
(iii) Does it match when rotated by 72°?
(iv) Draw it.

OR

(i) Show that the mirror image of the letter ‘E’ in a horizontal mirror is the same. 5
(ii) Give two other letters.

20. **Case study:** The lowest point on land is −400 m and the highest 8,848 m. 5

(i) Write as integers.
(ii) Find the difference in height.
(iii) Is the lowest point above sea level?
(iv) What is the midpoint?

OR

(i) Find the missing numbers: ___ + (−6) = −2; 7 + ___ = 0; (−9) + ___ = −15. 5
(ii) Explain each.

END OF QUESTION PAPER — Total: 40 Marks

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